0.1 Elementary Symmetric Functions

Definition 1 *A polynomial $P \in \mathbb{K}[X]$ is said to be completely decomposable over $\mathbb{K}$ if all its roots are in $\mathbb{K}$.*

Let $P = \sum_{k=0}^{n} a_k X^k \in \mathbb{K}[X]$ be completely decomposable over $\mathbb{K}$, and let $\alpha_1, \alpha_2, \ldots, \alpha_n \in \mathbb{K}$ be its roots. If a_n denotes its leading coefficient, P can be written as

$$P = a_n(X-\alpha_1)(X-\alpha_2)\ldots(X-\alpha_n) = a_n \prod_{k=1}^{n}(X-\alpha_k).$$

Example 2

1. *Let $P = X^3 - 2$. The polynomial P is completely decomposable over $\mathbb{C}$ but not over $\mathbb{R}$ since in $\mathbb{C}[X]$ we have:*

$$X^3 - 2 = (X - \sqrt[3]{2})(X - j\sqrt[3]{2})(X - j^2\sqrt[3]{2})$$

and in $\mathbb{R}[X]$ we have:

$$X^3 - 2 = (X - \sqrt[3]{2})(X^2 + X + (\sqrt[3]{2})^2).$$

2. *Let $P = X^4 - 3X^2 + 2$. The polynomial P is completely decomposable over $\mathbb{R}$ since $P = (X^2 - (2\sqrt{2}+3)X + \sqrt{2})(X^2 + (2\sqrt{2}+3)X + \sqrt{2})$, but not over $\mathbb{Q}$.*

Proposition 3 *Let $\mathbb{K}$ be an algebraically closed field, $P = \sum_{k=0}^{n} a_k X^k \in \mathbb{K}[X]$ (with $n \geq 1$), and $\alpha_1, \alpha_2, \ldots, \alpha_n$ the roots of P. Then we have*

$$a_{n-k} = a_n(-1)^k \sum_{1 \leq i_1 < i_2 < \cdots < i_k \leq n} \alpha_{i_1}\alpha_{i_2}\ldots\alpha_{i_k}.$$

Proof. Since $\mathbb{K}$ is algebraically closed, we have

$$P = \sum_{k=0}^{n} a_k X^k = a_n \prod_{k=1}^{n}(X-\alpha_k) \tag{1}$$

. Expanding the product $a_n \prod_{k=1}^{n}(X-\alpha_k)$ and equating the coefficients of the two members of the equality (1), we get the following identities

$$\sum_{k=1}^{n} \alpha_k = -\frac{a_{n-1}}{a_n},$$

$$\sum_{1 \leq j < k \leq n} \alpha_j \alpha_k = \frac{a_{n-2}}{a_n},$$

$$\vdots$$

$$\sum_{1\le i_1<i_2<\dots<i_k\le n} \alpha_{i_1}\alpha_{i_2}\dots\alpha_{i_k} = (-1)^k \frac{a_{n-k}}{a_n},$$

$$\vdots$$

$$\prod_{k=1}^{n} \alpha_k = (-1)^n \frac{a_0}{a_n}.$$

■

Definition 4 *Let $\alpha_1, \alpha_2, ..., \alpha_n$ be n elements of a field $\mathbb{K}$. The following expressions are called elementary symmetrical functions:*

$$\sigma_1 = \sum_{k=1}^{n} \alpha_k, \quad \sigma_2 = \sum_{1\le i_1<i_2\le n} \alpha_{i_1}\alpha_{i_2}, \dots, \quad \sigma_k = \sum_{1\le i_1<i_2<\dots<i_k\le n} \alpha_{i_1}\alpha_{i_2}\cdots\alpha_{i_k},$$

where $1 \le k \le n$.

For $n = 2$, we have two elementary functions:

$$\begin{cases} \sigma_1 = \alpha_1 + \alpha_2 \\ \sigma_2 = \alpha_1\alpha_2. \end{cases}$$

For $n = 3$, there are three elementary symmetrical functions:

Applications

1. Solve in $\mathbb{C}$ the following system $\begin{cases} x + y + z = 0 \\ xy + yz + xyz = -13 \\ xyz = -12. \end{cases}$

   Suppose that the unknowns x, y, z are the roots of a monic polynomial of degree 3 denoted P, to be determined. Then

   $$P = X^3 - \sigma_1 X^2 + \sigma_2 X + \sigma_3,$$

   where
   $$\begin{cases} \sigma_1 = x + y + z \\ \sigma_2 = xy + yz + xz \\ \sigma_3 = xyz. \end{cases}$$

   Thus, we have:

   $$P = X^3 - 13X + 12 = (X - 1)(X - 3)(X + 4).$$

   Then the set of solutions is

   $$S = \{(1, 3, -4), (1, 4, -3), (3, 1, -4), (3, -4, -1), (-4, 3, 1), (-4, 1, 3)\}.$$

2. Solve in $\mathbb{C}$ the system $\begin{cases} x + y + z = -2 \\ x^2 + y^2 + z^2 = 0 \\ x^3 + y^3 + z^3 = 1. \end{cases}$

Suppose that the unknowns x, y, z are the roots of P, a monic polynomial of degree 3.

Let

$$\begin{cases} S_1 = x + y + z \\ S_2 = x^2 + y^2 + z^2 \\ S_3 = x^3 + y^3 + z^3. \end{cases}$$

Then

$$\begin{cases} S_1 = \sigma_1 \\ S_2 = \sigma_1^2 - 2\sigma_2 \\ S_3 = \sigma_1^3 - 3\sigma_1\sigma_2 + 3\sigma_{3.} \end{cases}$$

This gives

$$\begin{cases} \sigma_1 = -2 \\ \sigma_1^2 - 2\sigma_2 = 0 \\ \sigma_1^3 - 3\sigma_1\sigma_2 + 3\sigma_3 = 1. \end{cases}$$

Thus,

$$\begin{cases} \sigma_1 = -2 \\ \sigma_2 = 2 \\ \sigma_3 = -1. \end{cases}$$

Hence, x, y, z are the solutions (up a permutation) of the polynomial

$$X^3 + 2X^2 + 2X + 1 = (X + 1)(X - j)(X - j^2).$$

Then the set of solutions is

$$T = \left\{(1, j, j^2), (1, j^2, j), (j, j^2, 1), (j^2, j, 1), (j, 1, j^2)), (j^2, 1, j)\right\}$$

0.2 Greatest Common Divisor and Least Common Multiple of Two Polynomials

0.2.1 Greatest Common Divisor of Two Polynomials

Let A and B be two nonzero polynomials in $\mathbb{K}[X]$. Let I be the set of common divisors of A and B. The set of degrees of polynomials in I is a finite subset of $\mathbb{N}_0$. Therefore, it admits a greatest element d. Hence, there exists in I a unique monic polynomial of degree d. This polynomial D is called *the greatest common divisor* of A and B, and is denoted as

$$\gcd(A, B) \text{ or } A \wedge B.$$

The greatest common divisor can be characterized as follows:

$$D=\gcd(A,B) \iff (D \text{ is monic and } \forall P\in\mathbb{K}[X], P\mid A \text{ and } P\mid B \text{ if and only if } P\mid D). \tag{2}$$

If A and B are two nonzero polynomials in $\mathbb{K}[X]$, the polynomial $\gcd(A,B)$ can be obtained by Euclid's algorithm. By this procedure, we can find polynomials U and V in $\mathbb{K}[X]$ such that

$$AU+BV=\gcd(A,B).$$

Indeed, to calculate the gcd of two polynomials A and B, we apply the Euclidean division of A by B (where B is nonzero and of degree less than or equal to that of A):

$$A=QB+R, \quad \deg\ R<\deg B.$$

The result is that

$$\gcd(A,B)=\gcd(B,R).$$

Consequently, if $R=0$, then

$$\gcd(A,B)=(\text{normalized form of } B).$$

If not, we repeat the procedure for the pair (B,R):

$$B=Q_1R+R_1, \qquad \deg R_1<\deg R,$$

and we have

$$\gcd(B,R)=\gcd(R,R_1).$$

We repeat the process until $R_n=0$.

$$\left|\begin{array}{cc} A=QB+R, & \deg R<\deg B \\ B=Q_1R+R_1, & \deg R_1<\deg R \\ R=Q_2R_1+R_2 & \deg R_2<\deg R_1 \\ R_1=Q_3R_2+R_3 & \deg R_3<\deg R_2 \\ \vdots & \vdots \\ R_{n-3}=Q_{n-1}R_{n-2}+R_{n-1} & \deg R_{n-1}<\deg R_{n-2} \\ R_{n-2}=Q_nR_{n-1}. & \end{array}\right.$$

Then,

$$\gcd(A,B)=\gcd(B,R)=\gcd(R,R_1)=\gcd(R_1,R_2)=\cdots=\gcd(R_{n-1},R_{n-2})=R_{n-1}.$$

The greatest common divisor is therefore the normalized last nonzero remainder.

Example 5 *In* $\mathbb{R}[X]$, *let*

$$A = X^5 + X^4 + 2X^3 + 2X^2 + X + 1, \quad B = 5X^4 + 4X^3 + 6X^2 + 4X + 1.$$

To find the gcd *of the polynomials* A *and* B*, we use the Euclidean algorithm.*

Dividing A by B, we get

$$R_1 = \frac{1}{5}X^4 + \frac{4}{5}X^3 + \frac{6}{5}X^2 + \frac{4}{5}X + 1.$$

Dividing B by R_1, we get

$$R_2 = -16X^3 - 24X^2 - 16X - 24.$$

Dividing R_1 by R_2, we get

$$R_3 = R_1 - \left(\frac{2}{5}X^4 + \frac{3}{10}X^3 + \frac{2}{5}X^2 + \frac{3}{10}X\right) = X^2 + 1.$$

Dividing R_2 by R_3, we get

$$R_4 = 0.$$

Then

$$\gcd(A, B) = X^2 + 1.$$

Here are some properties of the greatest common divisor of two polynomials.

Theorem 6 *Let* $A, B, C \in \mathbb{K}[X]$*, and* $\lambda \in \mathbb{K}^*$.

1. *Common divisors of* A *and* B *are exactly the divisors of* $\gcd(A, B)$.
2. *The polynomials* $\gcd(CA, CB)$ *and* $C \gcd(A, B)$ *are associates.*
3. $\gcd(\lambda A, \lambda B) = \gcd(A, B)$.

Proof.

1. If D is a common divisor of A and B, then, thanks to the characterization (2) of the gcd of two polynomials, D divides $\gcd(A, B)$. Conversely, let D_1 be a divisor of $\gcd(A, B)$. By the same characterization, D_1 divides A and B. Therefore, the common divisors of A and B are exactly the divisors of $\gcd(A, B)$.

2. The polynomials AC and BC are multiples of DC, so any divisor of DC divides AC and BC. Then, it divides $\gcd(AC, BC)$. Conversely, if Q divides AC and BC, then Q divides any combination $UAC+VBC, .(U,V \in \mathbb{K}[X])$. In particular, Q divides the combination

$$U_0AC + V_0BC = (U_0A + V_0B)C,$$

where $U_0, V_0 \in \mathbb{K}[X]$ satisfying

$$U_0A + V_0B = D.$$

Then, Q divides $C \gcd(A, B)$. Therefore $\gcd(AC, BC)$ and $C \gcd(A, B)$ have the same divisors and so they are associates.

3. Let $D_1 = \gcd(\lambda A, \lambda B)$ and $D_2 = \gcd(A, B)$. Since D_2 divides A and B, it also divides λA and λB and consequently, it divides D_1. Conversely, D_1 divides λA and λB, hence it divides A and B (since λ is a nonzero constant, it is invertible). Consequently, D_1 divides D_2. We conclude that D_1 and D_2 are associates. This implies $D_1 = \mu D_2$ for some nonzero constant μ. Since D_1 is monic, μ must be 1, and thus,

$$\gcd(\lambda A, \lambda B) = \gcd(A, B).$$

■

Definition 7 *Let $A, B \in \mathbb{K}[X]$. If $\gcd(A, B) = 1$, the polynomials A and B are said to be coprime (or relatively prime). In this case, the only common divisors of A and B are nonzero constant polynomials.*

Theorem 8 *Let $A, B \in \mathbb{K}[X]$. A and B are coprime if and only if there exist $U, V \in \mathbb{K}[X]$ such that*

$$UA + VB = 1.$$

Proof. Suppose $A, B \in \mathbb{K}[X]$ and there exist $U, V \in \mathbb{K}[X]$ such that

$$UA + VB = 1.$$

Let $D = \gcd(A, B)$. Since D divides A and D divides B, it follows that then D divides $UA + VB = 1$. This implies $D = 1$. Conversely, suppose $A, B \in \mathbb{K}[X]$ and $\gcd(A, B) = 1$. Euclid algorithm above ensures the existence of $U, V \in \mathbb{K}[X]$ such that $UA + VB = D = 1$. ■

Example 9 *In $\mathbb{R}[X]$, let $A = X^7 - X - 1$ and $B = X^5 - 1$. We will find two polynomials $U, V \in \mathbb{R}[X]$ such that*

$$UA + VB = 1.$$

Using Euclid's algorithm, we have

$$X^7 - X - 1 = (X^5 - 1)X^2 + X^2 - X - 1.$$

$$(X^5 - 1) = (X^2 - X - 1)(X^3 + X^2 + 2X + 3) + 5X + 2.$$

$$(X^2 - X - 1) = (5X + 2)(\frac{X}{5} - \frac{7}{25}) - \frac{11}{25}.$$

We then go back up the calculations. We successively get

$$-25(X^2 - X - 1) + (5X + 2)(5X - 7) = 11$$

$$-25(X^2 - X - 1) + (5X + 2)((X^5 - 1) - (X^2 - X - 1)(X^3 + X^2 + 2X + 3)) = 11.$$

Then

$$(X^2 - X - 1)(-5X^4 + 2X^3 - 3X^2 - X - 4) + (X^5 - 1)(5X - 7) = 11.$$

$$(-5X^4 + 2X^3 - 3X^2 - X - 4)(X^7 - X - 1) + (X^5 - 1)(5X^6 - 2X^5 + 3X^4 + X^3 + 4X^2 + 5X - 7) = 11.$$

Simply divide by 11 to obtain the polynomials U and V :

$$\begin{aligned} &\underbrace{\frac{(-5X^4 + 2X^3 - 3X^2 - X - 4)}{11}}_{U}\underbrace{(X^7 - X - 1)}_{A} \\ &+ \underbrace{\frac{(5X^6 - 2X^5 + 3X^4 + X^3 + 4X^2 + 5X - 7)}{11}}_{V}\underbrace{(X^5 - 1)}_{B} \\ &= 1. \end{aligned}$$

Proposition 10 *Let $A, B, D \in \mathbb{K}[X]$. Then*

$$\gcd(A, B) = D \textit{ if and only if } (\exists A_1, B_1 \in \mathbb{K}[X] : A = DA_1, B = DB_1 \textit{ and } \gcd(A_1, B_1) = 1). \tag{3}$$

Proof. Let $A, B, D \in K[X]$, and suppose $D = \gcd(A, B)$. Then D divides A and D divides B, which gives $A = DA_1$ and $B = DB_1$ for some $A_1, B_1 \in K[X]$. If we suppose $\gcd(A_1, B_1) = D_1 \neq 1$ (a non-constant polynomial), we get $A_1 = D_1A_2$ and $B_1 = D_1B_2$. Thus,

$$A = D_1A_2D \quad \text{and} \quad B = D_1B_2D,$$

which leads to $\gcd(A, B) \neq D$. Therefore, necessarily $\gcd(A_1, B_1) = 1$. Conversely, suppose that there exist $A_1, B_1 \in \mathbb{K}[X]$ such that $A = DA_1$, $B = DB_1$, and $\gcd(A_1, B_1) = 1$. It is clear that D divides A and D divides B, so D divides $\gcd(A, B)$. Since $\gcd(A_1, B_1) = 1$, there exist $U_0, V_0 \in \mathbb{K}[X]$ such that

$$A_1U_0 + B_1V_0 = 1.$$

Then,

$$\gcd(A, B) \text{ divides } A = DA_1 \text{ and } \gcd(A, B) \text{ divides } B = DB_1.$$

Thus,

$$\gcd(A, B) \text{ divides } DA_1U_0 + DB_1V_0 = D.$$

We conclude $\gcd(A, B) = D$, which completes the proof. ∎

Proposition: Let $A, B_1, B_2, \ldots, B_n \in \mathbb{K}[X]$. If

$$\gcd(A, B_k) = 1 \text{ for all } 1 \leq k \leq n,$$

then

$$\gcd(A, \prod_{k=1}^{n} B_k) = 1.$$

Proof. We prove the result for $n = 2$. Let $A, B_1, B_2 \in \mathbb{K}[X]$ such that $\gcd(A, B_1) = 1$ and $\gcd(A, B_2) = 1$. Then there exist $U_1, V_1, U_2, V_2 \in \mathbb{K}[X]$ such that

$$AU_1 + B_1V_1 = 1 \quad (1)$$

and

$$AU_2 + B_2V_2 = 1 \quad (2),$$

Multiplying (1) by (2) we get

$$AU_1AU_2 + AU_1B_2V_2 + B_1V_1AU_2 + B_1B_2V_1V_2 = 1,$$

which simplifies to

$$A(U_1AU_2 + U_1B_2V_2 + B_1V_1U_2) + B_1B_2(V_1V_2) = 1.$$

This is equivalent to

$$\gcd(A, B_1B_2) = 1.$$

∎

Theorem 11 (Gauss) *Let $A, B, C \in \mathbb{K}[X]$.*

If $\gcd(A, B) = 1$ and A divides (BC), then A divides C.

Proof. Since $\gcd(A, B) = 1$, according to Bezout's lemma there exist $U, V \in \mathbb{K}[X]$ such that
$$UA + VB = 1,$$
Since A divides BC, there exists $Q \in K[X]$ such that
$$BC = AQ.$$
Thus,
$$C = UAC + VBC = A(UC + QV).$$
Therefore, A divides C. ■

Proposition 12 (Euclid) *Let $P, Q, R \in \mathbb{K}[X]$. If P is irreducible and divides QR, then P divides Q or P divides R.*

Proof. Assume that P divides QR, and show that P divides Q or P divides R. Suppose that P does not divide Q and show that this implies P divides R. Since P does not divide Q and P is irreducible, we have $\gcd(P, Q) = 1$. If $\gcd(P, Q) \neq 1$, then there exists a polynomial $D \in K[X]$ such that D divides P. As P is irreducible, D must be 1 or associate with P. Since P does not divide Q, we must have $D = 1$, and thus $\gcd(P, Q) = 1$. Applying Gauss's lemma, since P divides QR and $\gcd(P, Q) = 1$, it follows that P divides R, as required. ■

0.2.2 Least Common Multiple of Two Polynomials

- Let A and B be two nonzero polynomials in $\mathbb{K}[X]$. The set of degrees of the common multiples to A and B is a subset of $\mathbb{N}$, and therefore, it admits a least element m. Consequently, there exists a unique monic polynomial M which is a common multiple of A and B with $\deg M = m$. This polynomial is called *least common multiple* of A and B, denoted by $M = \operatorname{lcm}(A, B)$ or $M = A \vee B$.
- The least common multiple of A and B can also be characterized by:

$$M = \operatorname{lcm}(A, B) \iff (M \text{ is monic and } \forall P \in K[X], (A \mid P \text{ and } B \mid P \implies M \mid P)). \tag{4}$$

▶ Recall that for any polynomial P we define $\operatorname{lcm}(P, 0) = 0$.

Properties of Least Common Multiple: The following properties are easy to show

1. For any two nonzero polynomials A and B in $\mathbb{K}[X]$, the set of common multiples to A and B exactly coincides with the set of multiples of their least common multiple.

2. Let A, B and C be three nonzero polynomials in $\mathbb{K}[X]$. Then the polynomials $\mathrm{lcm}(CA, CB)$ and $C\,\mathrm{lcm}(A, B)$ are associates.

3. Let A and B be two nonzero polynomials in $\mathbb{K}[X]$, let $D = \gcd(A, B)$ and $M = \mathrm{lcm}(A, B)$. Then the polynomials AB and MD are associates.